

A Dynamic Weighted Fusion Model for Multimodal Sentiment Analysis

Liang Yang¹, Xinhua Zhu^{1,2,*}, Siyang Li³

¹ School of Computer Science and Engineering, Guangxi Normal University, Guilin 541004, China

² Guangxi Key Laboratory of Multi-Source Information Mining and Security, Guilin 541004, China

³ UISEE (Shanghai) Automotive Technologies, Shanghai 201807, China

* Corresponding author: zxh429@263.net

Abstract

Multimodal sentiment analysis (MSA) tasks leverage diverse data sources, such as textual, acoustic, and visual data, to infer users' sentiment states. Previous research has mainly focused on capturing the differences and consistency of sentiment information between different modalities, emphasizing cross-modal interaction, while neglecting the in-depth exploration of sentiment information within individual modalities. Additionally, existing MSA methods rarely examine the contribution of each modality to model performance. To address these issues, our paper proposes a dynamic weighted fusion model for multimodal sentiment analysis. Specifically, we first design a Multi-Level Semantic Enhancement module (MLSE) for each single modality, aiming to extract semantic information from multiple perspectives and levels within the single mode. Subsequently, we utilize genetic algorithms to dynamically calculate the optimal weight parameters for each modality, thereby evaluating the impact of each modality on model performance. We conduct extensive experiments on three benchmark datasets (CMU-MOSI, CMU-MOSEI and CH-SIMS), and the results demonstrate that our proposed model outperforms state-of-the-art models across various metrics.

Keywords

Multimodal sentiment analysis; Multimodal interaction; Machine learning; Multimodal fusion;

1. Introduction

With the proliferation of information technology and social media, user-generated data displays increasingly diverse characteristics, providing a valuable resource for multimodal sentiment analysis [1], [2]. Multimodal sentiment analysis (MSA) integrates multiple sensory modalities, including textual, acoustic, and visual elements, to infer users' emotions and attitudes [3]. Traditional text-based sentiment analysis

一种用于多模态的动态加权融合模型

情感分析

梁阳¹, 朱新华^{1,2,*}, 李思阳³

¹ 广西师范大学计算机科学与工程学院, 桂林 541004, 中国 ² 广西多源信息挖掘与安全重点实验室, 桂林 541004, 中国 ³ 上海智驾汽车科技有限公司, 上海 201807, 中国 * 通讯作者
: zxh429@263.net

摘要

多模态情感分析 (MSA) 任务利用文本、声学 and 视觉等多种数据源来推断用户的情感状态。以往研究主要关注捕捉不同模态间情感信息的差异性与一致性, 强调跨模态交互, 却忽视了对单一模态内部情感信息的深入挖掘。此外, 现有 MSA 方法很少探究各模态对模型性能的贡献度。针对这些问题, 本文提出一种动态加权融合的多模态情感分析模型。具体而言, 我们首先为每个单模态设计了多层级语义增强模块 (MLSE), 旨在从单模态内部多视角、多层次提取语义信息。随后, 利用遗传算法动态计算各模态的最优权重参数 $\{v^*\}$, 从而评估各模态对模型性能的影响。我们在三个基准数据集 (CMU-MOSI、CMU-MOSEI 和 CH-SIMS) 上进行了大量实验, 结果表明, 我们提出的模型在各项指标上均优于当前最先进的模型。

关键词

多模态情感分析; 多模态交互; 机器学习; 多模态融合;

1. 引言

随着信息技术和社交媒体的普及, 用户生成的数据呈现出日益多样化的特征, 为多模态情感分析提供了宝贵的资源[1]、[2]。多模态情感分析 (MSA) 融合了包括文本、听觉和视觉元素在内的多种感官模态, 以推断用户的情感和态度[3]。传统的基于文本的情感分析

primarily relies on identifying emotional vocabulary and phrases in text to infer user emotions [4], [5]. However, human emotions are often complex, involving speech intonation, facial expressions, gestures, and other nonverbal cues. Traditional text-based sentiment analysis struggles to address the intricacies of this data, and sentiment analysis research is gradually shifting from unimodal to multimodal approaches [6]. By incorporating these diverse emotional expressions, multimodal sentiment analysis can more accurately and comprehensively capture user emotions, thereby improving the reliability and effectiveness of sentiment prediction.

Previous works have made significant progress in exploring the complementarity between multimodal data [7], [8], [9]. However, prior research on multimodal sentiment analysis (MSA) has predominantly focused on capturing the differences and consistency of emotional information across various modalities, emphasizing the interaction between these modalities [9], [10]. This has often come at the expense of a thorough exploration of emotional information within a single mode. Additionally, current MSA methods seldom address the specific contributions of individual modes to the overall model performance. When information from different modes is inconsistent or mismatched, accurately and effectively identifying user emotions poses a significant challenge [10]. For instance, as shown in Fig 1, the text is expressed as “*And um while the tracker made the film look real nice and cute*”. From words “*real nice*” and “*cute*”, we can extract positive and positive emotions, but the facial expression can be inferred as neutral or even negative, and the corresponding emotion label is negative. In this case, relying on cross modal interaction alone is not enough to compensate for the loss of potential information or misleading clues introduced by a single mode, and the inconsistency of modal information poses a complex challenge to emotion recognition. Therefore, effectively integrating information from various modes, exploring the weight ratio between each mode, and more accurately identifying the emotional state of users have become urgent issues that need to be addressed.

To address the issue of multimodal sentiment representation mentioned above, we propose a dynamic weighted fusion framework for multimodal sentiment analysis. Specifically, we design a Multi-Level Semantic Enhancement module (MLSE), which leverages convolutional neural networks [11] and multi-head attention mechanisms [12] to capture abstract information and contextual relationships within each modality, thereby enhancing the representational capacity of each modality at multiple levels. Then, we use genetic algorithms to optimize the weight parameters of each modality, maximizing the contribution of each modality's emotional features to the overall model performance. Through the iterative optimization process of the genetic algorithm, the model can automatically adjust the modal weights to flexibly adapt to changes in modal information across different environments, calculating the most

主要依赖于识别文本中的情感词汇和短语来推断用户情绪[4]、[5]。然而，人类情感往往复杂多样，涉及语音语调、面部表情、手势及其他非语言线索。传统的基于文本的情感分析难以处理此类数据的复杂性，因此情感分析研究正逐渐从单模态转向多模态方法[6]。通过整合这些多样化的情感表达，多模态情感分析能够更准确、更全面地捕捉用户情绪，从而提升情感预测的可靠性和有效性。

先前的研究在多模态数据互补性探索方面取得了显著进展[7]、[8]、[9]。然而，现有的多模态情感分析研究主要聚焦于捕捉不同模态间情感信息的差异性与一致性，强调模态间的交互作用[9]、[10]，这往往以牺牲对单一模态内部情感信息的深入挖掘为代价。此外，当前的多模态情感分析方法很少探讨各模态对模型整体性能的具体贡献。当不同模态的信息存在不一致或失配时，准确有效地识别用户情感面临重大挑战[10]。例如，如图1所示，文本内容表达为“*And um while the tracker made the film look real nice and cute*”。从词语“*real nice*”和“*cute*”中可提取出积极情感，但面部表情可推断为中性甚至消极，而对应的情感标签实为负面。在这种情况下，仅依赖跨模态交互不足以弥补单一模态可能造成的信息潜在损失或误导性线索，且模态信息的不一致性为情感识别带来了复杂挑战。因此，如何有效整合多模态信息、探索各模态间的权重配比，从而更精准地识别用户情感状态，已成为亟待解决的关键问题。

针对上述多模态情感表征问题，我们提出了一种动态加权融合框架用于多模态情感分析。具体而言，我们设计了多层级语义增强模块（MLSE），该模块利用卷积神经网络[11]与多头注意力机制[12]捕捉各模态内部的抽象信息及上下文关联，从而在多个层次提升各模态的表征能力。随后，我们采用遗传算法对各模态的权重参数进行优化，最大化各模态情感特征对整体模型性能的贡献。通过遗传算法的迭代优化过程，模型能自动调整模态权重，灵活适应不同环境下模态信息的变化，计算出最

suitable weights for each modality. Lastly, we utilize weighted fusion strategy to derive the final emotional prediction outcomes. This approach enhances the semantic information of each modality at multiple levels by deeply mining individual modalities. Additionally, when modal information is inconsistent or mismatched, the model can calculate the similarity between each modality and the true sentiment labels, dynamically adjusting each modality's weight ratio and contribution. This provides innovative solutions for the field of multimodal sentiment analysis. Our primary contributions can be outlined as follows:

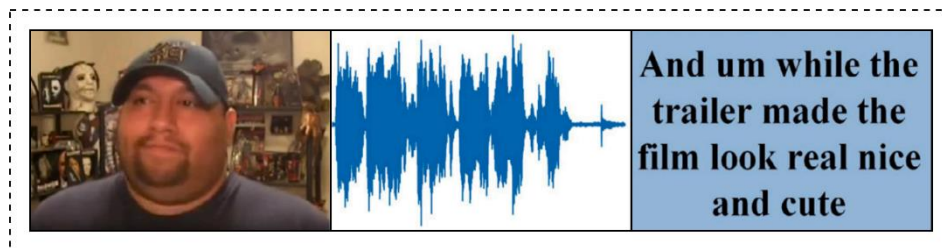


Fig 1. Inconsistent emotions expressed between modalities

1. We have designed a multi-level semantic enhancement module that uses convolutional neural networks and self-attention mechanisms to capture abstract information and contextual relationships across modalities, enhancing the representational capabilities of each modality at multiple levels.

2. We introduce a genetic algorithm suitable for multimodal sentiment analysis, which dynamically evolves the weight parameters of each modality in different contexts to improve the accuracy and stability of sentiment analysis.

3. Extensive experiments conducted on three benchmark datasets demonstrate that our proposed method exhibits superior performance compared to existing advanced models.

2. Related work

2.1 Multimodal Sentiment Analysis

In recent years, the field of multimodal emotion analysis has attracted more and more attention [13]. Multimodal emotion analysis combines information from different sensory modes, such as text, voice, image, etc., to infer and understand users' emotions, attitudes and emotional tendencies [14], [15]. In comparison to traditional sentiment analysis methods based on a single mode, multimodal sentiment analysis can capture and comprehend users' emotions more comprehensively

错误!未找到引用源。 . In their early work, Zadeh et al. [7] proposed to integrate all modal information into high-dimensional tensors by using triple Cartesian product operation

为每种模态分配合适的权重。最后，我们采用加权融合策略得出最终的情感预测结果。该方法通过深度挖掘各独立模态，在多个层次上增强了每种模态的语义信息。此外，当模态信息不一致或不匹配时，模型能够计算各模态与真实情感标签之间的相似度，动态调整各模态的权重比例及贡献度。这为多模态情感分析领域提供了创新性解决方案。我们的主要贡献可概括如下：

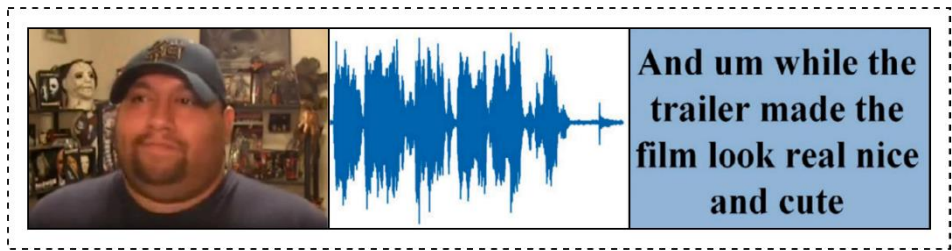


图1. 模态间表达的情感不一致

1. 我们设计了一个多级语义增强模块，该模块利用卷积神经网络和自注意力机制来捕捉跨模态的抽象信息和上下文关系，从而在多个层次上增强每种模态的表征能力。
2. 我们引入了一种适用于多模态情感分析的遗传算法，该算法能动态演化不同情境下各模态的权重参数，以提升情感分析的准确性与稳定性。
3. 在三个基准数据集上进行的大量实验表明，我们提出的方法相较于现有先进模型展现出更优越的性能。

2. 相关工作

2.1 多模态情感分析

近年来，多模态情感分析领域受到越来越多的关注[13]。多模态情感分析结合了来自不同感官模式的信息，如文本、语音、图像等，以推断和理解用户的情绪、态度及情感倾向[14]、[15]。相较于基于单一模式的传统情感分析方法，多模态情感分析能更全面地捕捉和理解用户的情感 $\{v^*\}$ 。

在他们的早期工作中，Zadeh等人[7]提出通过使用三重笛卡尔积运算，将所有模态信息整合到高维张量中。

and LSTM network [17], but this method has high computational complexity. In order to simplify this complex calculation, Liu et al. [8] proposed a low rank tensor fusion network (LMF) to improve the efficiency of data processing by reducing the parameters of the low rank tensor. In order to strengthen the integration of cross modes, Wang et al. [18] proposed a hierarchical fusion network (Raven) with both local fusion and global fusion. This network captures the fine-grained relationship between modes through the sliding window mechanism while maintaining computational efficiency. With the proposal of transformer [12], more and more researchers apply it to MSA tasks. Tsai et al. [19] utilized directional pairwise cross-modal attention mechanisms to process information interaction between different modalities, and proposed Multimodal Transformer (MulT). MulT can effectively handle unaligned multimodal language sequences and learn the correlations and interactions between different modalities. The TEDT model proposed by Wang et al. [20] introduces a multimodal encoding decoding network, which takes the text mode as the main information to improve the performance of the model. In recent research, Zhang et al. [21] proposed the ICDN model to solve the time series alignment and missing modal information by introducing a cross-modal converter. Wang et al. [22] proposed a text enhancement converter fusion network (TETFN), which focuses on learning text oriented pairwise cross modal mapping to obtain a unified and robust multimodal representation. Yao et al. [23] proposed a multi-modal feature decoupling strategy that categorizes sentiment features into common and private features (FDR-MSA), the important information in the decoupling process is retained and the influence of redundant data is reduced by introducing a dual feature reconstruction mechanism composed of unimodal feature reconstruction (UFR) and multi-modal feature reconstruction (MFR).

2.2 Modalities fusion strategy

Effectively integrating features from different modalities is critical in multimodal sentiment analysis [24]. The current multimodal fusion strategies are primarily categorized into two groups: traditional methods and deep learning-based methods. Illustrated in Fig 2, traditional methods comprise early fusion and late fusion, while deep learning-based methods are further classified into attention-based fusion and attention-unrelated fusion.

In early fusion strategies, features from different modalities are merged at the input layer, streamlining the feature fusion process. Self-MM [25] designed a label generation module based on a self-supervised learning strategy, dividing multimodal sentiment analysis tasks into multimodal tasks and single modal subtasks. Joint training is implemented on both multimodal and single modal tasks to learn the consistencies and discrepancies between modalities. Late fusion, commonly referred

以及LSTM网络[17]，但该方法计算复杂度较高。为简化这一复杂计算，Liu等人[8]提出了低秩张量融合网络（LMF），通过降低低秩张量的参数来提升数据处理效率。为加强跨模态的融合，Wang等人[18]提出了一种兼具局部融合与全局融合的分层融合网络（Raven）。该网络通过滑动窗口机制捕捉模态间的细粒度关系，同时保持计算效率。随着Transformer[12]的提出，越来越多的研究者将其应用于多模态情感分析任务。Tsai等人[19]利用定向成对跨模态注意力机制处理不同模态间的信息交互，提出了多模态Transformer（MulT）。MulT能有效处理未对齐的多模态语言序列，并学习不同模态间的关联与交互。Wang等人[20]提出的TEDT模型引入了多模态编码解码网络，以文本模态为主要信息来提升模型性能。在近期研究中，Zhang等人[21]提出ICDN模型，通过引入跨模态转换器解决时间序列对齐与模态信息缺失问题。Wang等人[22]提出文本增强转换器融合网络（TETFN），专注于学习以文本为导向的成对跨模态映射，以获得统一且鲁棒的多模态表示。Yao等人[23]提出一种多模态特征解耦策略，将情感特征划分为公共特征与私有特征（FDR-MSA），通过引入由单模态特征重建（UFR）和多模态特征重建（MFR）构成的双特征重建机制，保留解耦过程中的重要信息并减少冗余数据的影响。

2.2 模态融合策略

在多模态情感分析中，有效整合不同模态的特征至关重要[24]。当前的多模态融合策略主要分为两类：传统方法与基于深度学习的方法。如图2所示，传统方法包含早期融合与晚期融合，而基于深度学习的方法则进一步细分为基于注意力的融合与非注意力融合。

在早期融合策略中，不同模态的特征在输入层进行合并，从而简化了特征融合过程。Self-MM [25] 基于自监督学习策略设计了一个标签生成模块，将多模态情感分析任务划分为多模态任务和单模态子任务。通过对多模态和单模态任务进行联合训练，学习模态间的一致性和差异性。晚期融合，通常被称为

to as decision-level fusion, entails the process of learning by independently introducing features from various modalities into their designated models. Subsequently, these modal-specific features are integrated at a more advanced stage within the model. The primary methods encompassed in this approach are averaging [26], majority voting [27], weighted sum [28], etc. Late fusion tends to lack consideration for the interaction information between different modalities, which may lead to overlooking the complementarity and correlation between these modalities. Attention-based fusion strategies typically utilize various attention mechanisms to achieve information exchange between or within modalities. MARN [29] utilized the Multi-attention Block (MAB) to discover interactions between modalities and store them within the hybrid memory of a recurrent component called the Long-Short Term Hybrid Memory (LSTHM). Ou et al. [30] proposed a multimodal local global attention network (MMLGAN) for emotional video content analysis extends the attention mechanism to multi-level fusion and designs a multimodal fusion unit to obtain a global representation of emotional videos. Attention-based methods focus more on the correlation and consistency between different modalities, which may lead to the model ignoring the differences between modalities when weighting different modal features.

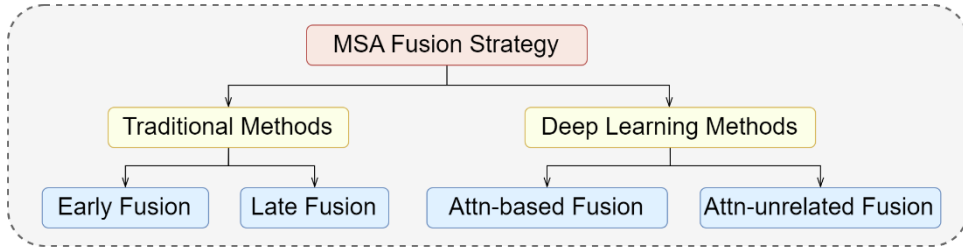


Fig 2. MSA modalities fusion strategy

3. Methodology

This section first introduces the task definition, and then details the proposed model framework. Figure 2 shows the overall structure of the proposed model. We will elaborate on the proposed model from four aspects: feature extraction, feature enhancement, weight calculation, and weighted fusion.

作为决策级融合，其过程涉及将来自不同模态的特征独立引入到其指定的模型中学习。随后，这些特定于模态的特征在模型内更高级的阶段进行整合。该方法主要涵盖平均[26]、多数投票[27]、加权求和[28]等方式。晚期融合往往缺乏对不同模态间交互信息的考量，可能导致忽略这些模态之间的互补性与相关性。基于注意力的融合策略通常利用各种注意力机制来实现模态间或模态内的信息交换。MARN[29]使用多注意力块（MAB）来发现模态间的交互，并将其存储在名为长短时混合记忆（LSTHM）的循环组件的混合记忆中。Ou等人[30]提出了一种用于情感视频内容分析的多模态局部全局注意力网络（MMLGAN），将注意力机制扩展到多级融合，并设计了一个多模态融合单元以获得情感视频的全局表示。基于注意力的方法更关注不同模态间的相关性和一致性，这可能导致模型在对不同模态特征进行加权时忽略模态间的差异。

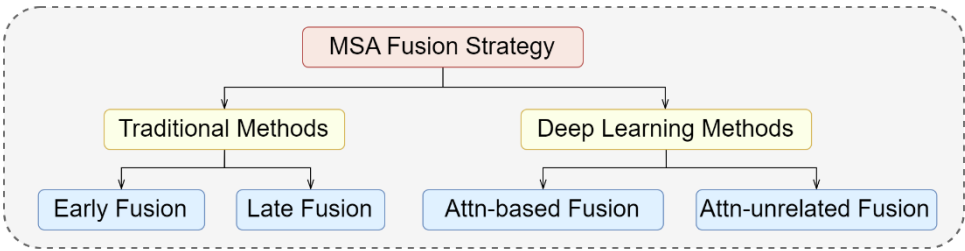


图2. MSA模态融合策略

3. 方法论

本节首先介绍任务定义，随后详细阐述所提出的模型框架。图2展示了所提出模型的整体结构。我们将从特征提取、特征增强、权重计算及加权融合四个方面详细说明该模型。

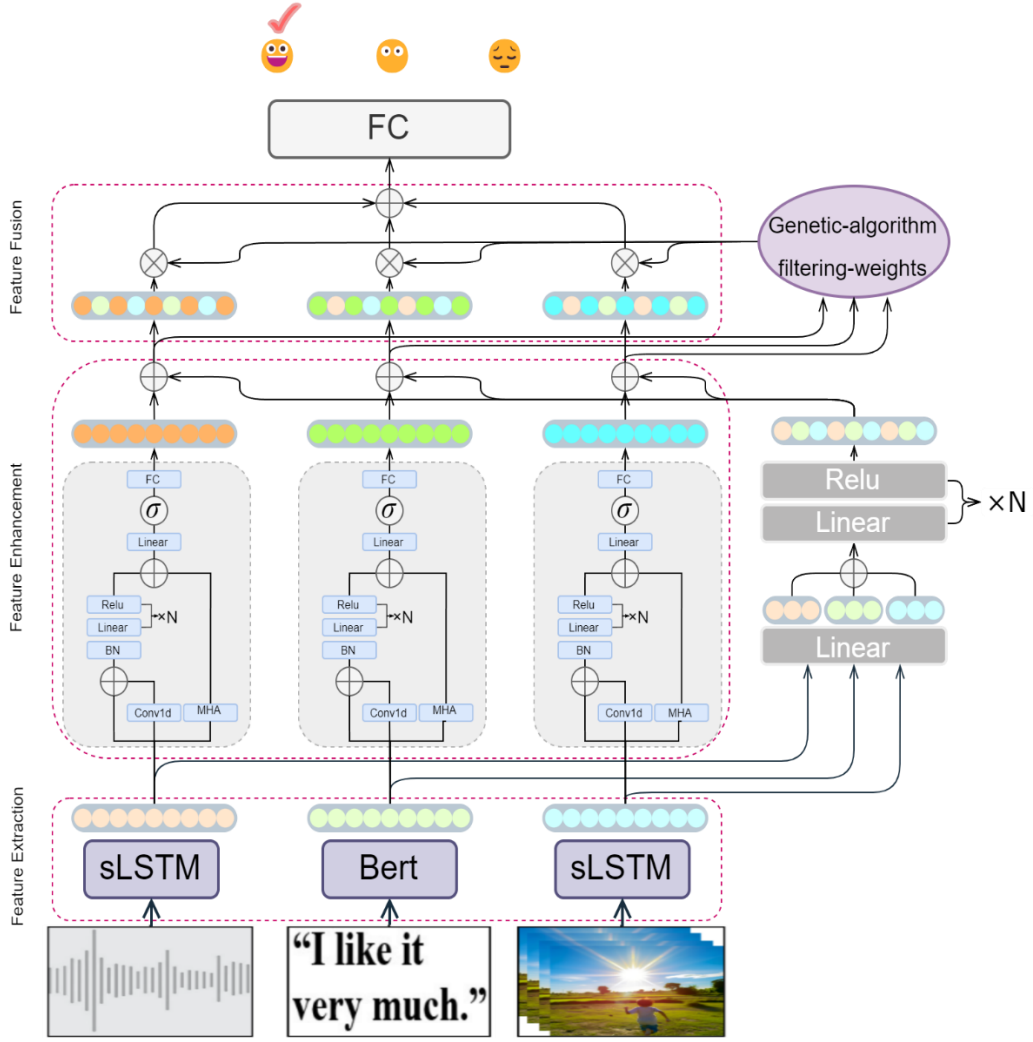


Fig 3. Overall architecture of the proposed Model

3.1 Task definition

The task of Multimodal Sentiment Analysis (MSA) aims to deduce and identify the emotional polarity conveyed by users through the analysis of various modalities of information. The benchmark dataset used in this study divides video segments into text (t), audio (a), and visual (v) signals, and these feature signals of different modalities will serve as inputs to the model. The symbol is represented as $U_m \in \mathbb{R}^{l_m \times d_m}$, where l_m denotes the sequence length, d_m denotes the feature dimension, $m \in \{t, a, v\}$. Correspondingly, the model outputs an emotion intensity result Y^p . Specifically, $Y^p > 0$ indicates positive sentiments, $Y^p < 0$ indicates negative sentiments, and $Y^p = 0$ indicates neutral sentiments.

3.2 Feature preprocessing and extraction

Following previous research [9], [25], the pre-trained BERT [31] is proficient in capturing bidirectional contextual information and rich semantic relationships

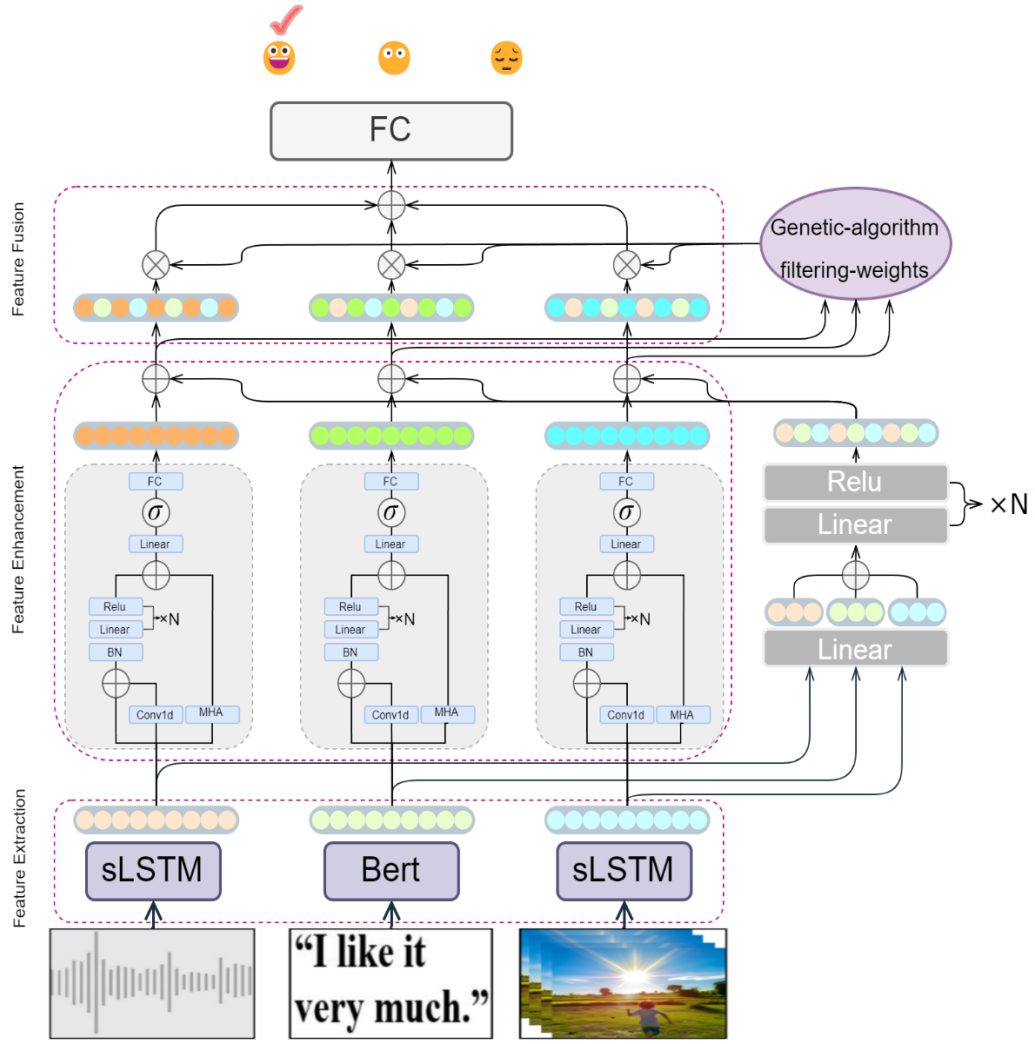


图3. 所提出模型的整体架构

3.1 任务定义

多模态情感分析（MSA）的任务旨在通过分析多种信息模态，推断并识别用户所传达的情感极性。本研究采用的基准数据集将视频片段划分为文本（t）、音频（a）和视觉（v）信号，这些不同模态的特征信号将作为模型的输入。符号表示为 $U_m \in \mathbb{R}^{l_m \times d_m}$ ，其中 l_m 表示序列长度， d_m 表示特征维度， $m \in \{t, a, v\}$ 。相应地，模型输出情感强度结果 Y^p 。具体而言， $Y^p > 0$ 表示积极情感， $Y^p < 0$ 表示消极情感， $Y^p = 0$ 表示中性情感。

3.2 特征预处理与提取

根据先前的研究[9]、[25]，预训练的BERT[31]擅长捕捉双向上下文信息及丰富的语义关系。

between different vocabulary, generating sentence representations imbued with rich sentiment information. Therefore, we select the BERT model which consisting of 12 stacked transformers as the text encoder, and the first word vector of the last layer of BERT to represent the output features. The representation of the resulting text modality is

$$F_t = BERT(U_t; \theta_t^{bert}) \in \mathbb{R}^{d_t}$$

where U_t represents the original text data, θ_{bert} denotes all learnable parameters of the pre-trained BERT, and d_t is the feature dimensions.

For visual modality, the three datasets use OpenFace2.0 toolkit [32] for initial video processing, including extracting facial action units, facial markers, head posture, line of sight tracking, and HOG functions [38], [40]. Finally, 709-dimensional frame-level visual features are extracted by CH-SIMS [33], 35-dimensional by CMU-MOSI [34] and CMU-MOSEI [35]. Then we use LSTM network [36] to process the preprocessed visual features U_v which provided by these three datasets. Long-Short-Term-Memory (LSTM) is a commonly used variant of Recurrent Neural Network (RNN) [37], which has stronger memory ability, solves the problem of gradient vanishing, and is more suitable for processing long sequence data. We use the hidden state of the final time step of LSTM to represent the visual modality, which contains information about the entire input sequence and can be used for the final prediction or further processing of the network.

$$F_v = LSTM(U_v; \theta_v^{lstm}) \in \mathbb{R}^{d_v}$$

For audio modality, the datasets CMU-MOSI and CMU-MOSEI use the speech processing tool COVAREP [39] to extract feature parameters, such as Mel-frequency cepstral coefficients (MFCCs), volume, glottal source, and other features related to emotion and intonation. These feature parameters can be obtained through the CMU Multimodal SDK. The dataset CH-SIMS uses the LibROSA speech toolkit [41] to extract 33-dimensional acoustic features at 22050Hz with default parameters, which are related to emotions and mood [42]. Consistent with visual feature processing techniques, we have also incorporated LSTM to extract and analyze the preprocessed acoustic features,

$$F_a = LSTM(U_a; \theta_a^{lstm}) \in \mathbb{R}^{d_a}$$

where U_v and U_a denote the preprocessed audio and visual features respectively, and θ_m^{lstm} represents all learnable parameters in LSTM, $m \in \{v, a\}$.

3.3 Multi-Level Semantic Enhancement Fusion module

This section discusses the multi-level semantic enhancement module of the modality proposed in this study. After feature extraction in the previous section, we use the

在不同词汇之间，生成蕴含丰富情感信息的句子表征。因此，我们选择由12层堆叠Transformer构成的BERT模型作为文本编码器，并采用BERT最后一层的首个词向量来表征输出特征。所得文本模态的表示为

$$F_t = BERT(U_t; \theta_t^{bert}) \in \mathbb{R}^{d_t}$$

其中 U_t 代表原始文本数据， θ_{bert} 表示预训练 BERT 的所有可学习参数，而 d_t 是特征维度。

对于视觉模态，这三个数据集使用OpenFace2.0工具包[32]进行初始视频处理，包括提取面部动作单元、面部标记点和头部姿态。

视线跟踪和HOG函数[38]、[40]。最后，通过CH-SIMS[33]提取709维帧级视觉特征，通过CMU-MOSI[34]和CMU-MOSEI[35]提取35维特征。随后，我们使用LSTM网络[36]处理这三个数据集提供的预处理视觉特征 U_v 。长短期记忆网络（LSTM）是循环神经网络（RNN）[37]的一种常用变体，具有更强的记忆能力，能解决梯度消失问题，更适合处理长序列数据。我们使用LSTM最后一个时间步的隐藏状态来表示视觉模态，该状态包含整个输入序列的信息，可用于网络的最终预测或进一步处理。

$$F_v = LSTM(U_v; \theta_v^{lstm}) \in \mathbb{R}^{d_v}$$

对于音频模态，数据集CMU-MOSI和CMU-MOSEI使用语音处理工具COVAREP[39]提取特征参数，例如梅尔频率倒谱系数（MFCCs）、音量、声门源以及其他与情感和语调相关的特征。这些特征参数可通过CMU多模态SDK获取。数据集CH-SIMS使用LibROSA语音工具包[41]以默认参数在22050Hz下提取33维声学特征，这些特征与情绪和情感状态相关[42]。与视觉特征处理技术保持一致，我们也采用LSTM来提取和分析预处理后的声学特征。

$$F_a = LSTM(U_a; \theta_a^{lstm}) \in \mathbb{R}^{d_a}$$

其中 U_v 和 U_a 分别表示预处理后的音频和视觉特征， θ_m^{lstm} 代表 LSTM 中所有可学习的参数 $m \in \{v, a\}$ 。

3.3 多级语义增强融合模块

本节讨论本研究中提出的模态多级语义增强模块。在上一节的特征提取之后，我们使用

Multi-Level Semantic Enhancement Fusion (MLSE) module to further enhance the semantic information of modal features from multi-levels. Specifically, we replicate the feature F_m obtained in the previous step into three duplicates. The first duplicate retains the original form, and the second undergoes processing through a one-dimensional convolutional network [11] to amplify the local representation of the modality. Conv1D layers are valuable in extracting important local patterns in sequential data, providing flexibility and efficiency in effectively modeling one-dimensional data structures. Next, we concatenate them and pass them through a Batch Norm layer, which helps improve the model's performance, accelerates the training process, and enhances the generalization ability of the model. Then, after passing through N linear layers, the output is obtained. The specific calculation process is shown in formulas:

$$F'_m = \text{Conv1D}(F_m, k_m)$$

$$F''_m = \text{BN}(F_m \oplus F'_m)$$

$$F_m^{\text{local}} = N \times \text{RELU}(\text{Linear}(F''_m); \theta_m)$$

where k_m denotes the kernel size for each modality, $\text{BN}(\cdot)$ denotes BatchNorm layer, θ_m denotes learnable parameters, $m \in \{t, v, a\}$.

We further utilize the multi-head attention to enhance the global semantic information of the third duplicate. Attention is proficient in capturing the dependency relationship between any two positions (tokens) in the input sequence, regardless of the distance between the two positions, which helps the model better understand the global structure and relationships in the input sequence,

$$F_m^{\text{global}} = \text{MultiHead}(F_m, F_m, F_m) = \text{Concat}(\text{head}_1, \text{head}_2, \dots, \text{head}_k)W^o$$

$$\text{head}_i = \text{Attention}(F_m W_i^q, F_m W_i^k, F_m W_i^v)$$

where W^o , W_i^q , W_i^k and W_i^v are learnable weight matrices.

Finally, we use a gated network to fuse multi-level semantic information:

$$F_m^{\text{output}} = \text{FC} \left(\sigma(F_m^{\text{local}} \oplus F_m^{\text{global}}) \right)$$

where $\sigma(\cdot)$ and FC denote sigmoid activation functions and fully-connected layers respectively.

多级语义增强融合（MLSE）模块，旨在从多层次进一步强化模态特征的语义信息。具体而言，我们将前一步得到的特征 F_m 复制为三份副本。第一份保留原始形式，第二份通过一维卷积网络[11]进行处理，以增强模态的局部表征能力。Conv1D层在提取序列数据中的重要局部模式方面具有显著价值，能够灵活高效地对一维数据结构进行有效建模。接着，我们将二者拼接后送入批归一化层，这有助于提升模型性能、加速训练过程并增强模型的泛化能力。随后，经过N个线性层处理，最终获得输出。具体计算过程如公式所示：

$$F'_m = \text{Conv1D}(F_m, k_m)$$

$$F''_m = \text{BN}(F_m \oplus F'_m)$$

$$F_m^{\text{local}} = N \times \text{RELU}(\text{Linear}(F''_m); \theta_m)$$

其中 k_m 表示每个模态的核大小， $\text{BN}(\cdot)$ 表示批归一化层， θ_m 表示可学习参数， $m \in \{t, v, a\}$ 。

我们进一步利用多头注意力机制来增强第三个副本的全局语义信息。注意力机制擅长捕捉输入序列中任意两个位置（标记）之间的依赖关系，无论这两个位置之间的距离如何，这有助于模型更好地理解输入序列的全局结构和关联，

$$F_m^{\text{global}} = \text{MultiHead}(F_m, F_m, F_m) = \text{Concat}(\text{head}_1, \text{head}_2, \dots, \text{head}_k) W^o$$

$$\text{head}_i = \text{Attention}(F_m W_i^q, F_m W_i^k, F_m W_i^v)$$

其中 W^o 、 W_i^q 、 W_i^k 和 W_i^v 是可学习的权重矩阵。

最后，我们使用一个门控网络来融合多层次的语义信息：

$$F_m^{\text{output}} = \text{FC}(\sigma(F_m^{\text{local}} \oplus F_m^{\text{global}}))$$

其中 $\sigma(\cdot)$ 和 FC 分别表示 sigmoid 激活函数和全连接层。

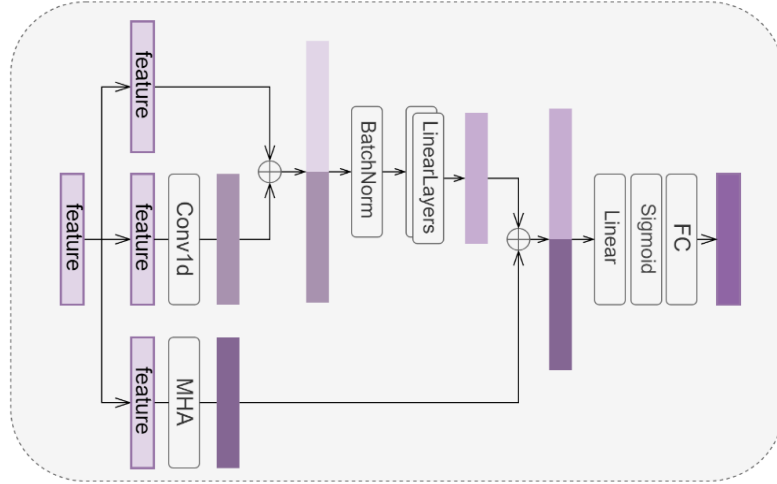


Fig 4. The Multi-Level Semantic Fusion Module

3.4 Dynamic weight evolution

Genetic algorithms (GA) are optimization algorithms inspired by biological evolution, used to search and optimize the solution space of problems.

Algorithm 1 The general process of genetic algorithm

- 1: Initialize population
 - 2: **while** termination condition not met **do**
 - 3: Evaluate individuals
 - 4: Select parents
 - 5: Recombine selected parents
 - 6: Mutate offspring
 - 7: Replace population with new generation
 - 8: **end while**
 - 9: **return** Best individual
-

Algorithm 2 The Evaluate function of genetic algorithm

- 1: **function** Evaluate fitness (weights, data)
 - 2: text data, audio data, visual data, labels $\leftarrow$ data
 - 3: text score $\leftarrow$ weights [0] $\times$ text data
 - 4: audio score $\leftarrow$ weights [1] $\times$ audio data
 - 5: visual score $\leftarrow$ weights [2] $\times$ visual data
 - 6: layer $\leftarrow$ a linear layer without bias
 - 7: text score, audio score, visual score $\leftarrow$ apply layer
 - 8: labels $\leftarrow$ extract numpy labels
 - 9: **return** sum of mean absolute errors between scores and labels
 - 10: **end function**
-

3.5 Feature Fusion

3.6

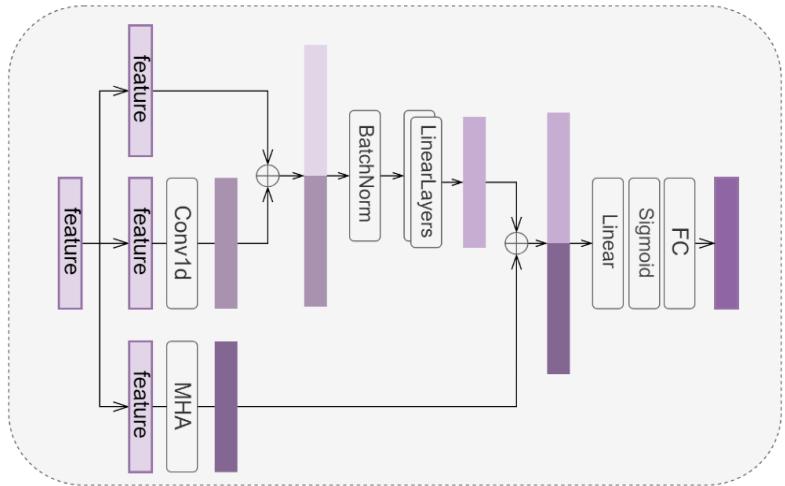


图4. 多级语义融合模块

3.4 动态权重演化

遗传算法（GA）是受生物进化启发的优化算法，
用于搜索和优化问题的解空间。

Algorithm 1 The general process of genetic algorithm

- 1: Initialize population
 - 2: **while** termination condition not met **do**
 - 3: Evaluate individuals
 - 4: Select parents
 - 5: Recombine selected parents
 - 6: Mutate offspring
 - 7: Replace population with new generation
 - 8: **end while**
 - 9: **return** Best individual
-

Algorithm 2 The Evaluate function of genetic algorithm

- 1: **function** Evaluate fitness (weights, data)
 - 2: text data, audio data, visual data, labels $\leftarrow$ data
 - 3: text score $\leftarrow$ weights [0] $\times$ text data
 - 4: audio score $\leftarrow$ weights [1] $\times$ audio data
 - 5: visual score $\leftarrow$ weights [2] $\times$ visual data
 - 6: layer $\leftarrow$ a linear layer without bias
 - 7: text score, audio score, visual score $\leftarrow$ apply layer
 - 8: labels $\leftarrow$ extract numpy labels
 - 9: **return** sum of mean absolute errors between scores and labels
 - 10: **end function**
-

3.5 特征融合

3.6

4. Experiments

In this section, we empirically evaluated the performance of three commonly used public dataset models in multimodal sentiment analysis tasks. Specifically, this section focuses on the dataset used, evaluation metrics, baseline methods, and experimental details.

4.1 Datasets

This study used three common benchmark datasets, CMU-MOSI [34], CMU-MOSEI [35], and CH-SIMS [33], which are widely used in MSA tasks, to train and test our proposed model. The first two datasets are in English and the last one is in Chinese. The number of emotion categories for each dataset is shown in Table 1, and we have briefly introduced these datasets.

Table 1 Dataset statistics in MOSI, MOSEI, and SIMS.

Dataset	Train	Valid	Test	All
CMU-MOSI	1284	229	686	2199
CMU-MOSEI	16326	1871	4659	22856
CH-SIMS	1368	456	457	2281

CMU-MOSI: The CMU-MOSI dataset is a multimodal dataset used for sentiment analysis and sentiment intensity prediction. This dataset mainly includes 93 short monologue video clips extracted from 2199 YouTube movie review videos. It covers facial expressions, speech, text, and contextual information. Each video clip is labeled with emotional labels, ranging from -3 (highly negative) to 3 (highly positive), used to indicate emotional tendencies and intensity.

CMU-MOSEI: This is an enhanced version of CMU-MOSI, significantly expanding its scale and scope. The dataset contains 22856 annotated video clips covering 250 different topics. This dataset has the same tag values as the CMU-MOSI dataset.

CH-SIMS: The CH-SIMS dataset is a Chinese MSA dataset that collects a total of 60 original videos from movies, TV shows, and variety shows. These videos are split into 2281 video clips, providing comprehensive multimodal annotations and individual annotations for each modality. Each sample in the dataset was assigned an emotional score ranging from -1 (indicating strong negativity) to 1 (indicating strong positivity).

4.2 Baselines

4. 实验

在本节中，我们通过实验评估了三种常用公共数据集模型在多模态情感分析任务中的性能。具体而言，本节重点介绍了所使用的数据集、评估指标、基线方法以及实验细节。

4.1 数据集

本研究使用了三个常用的基准数据集——CMU-MOSI [34]、CMU-MOSEI [35] 和 CH-SIMS [33]——来训练和测试我们提出的模型，这些数据集在MSA任务中被广泛使用。前两个数据集为英文，最后一个为中文。各数据集的情感类别数量如表1所示，我们已对这些数据集进行了简要介绍。

表1 MOSI、MOSEI和SIMS数据集统计信息。

Dataset	Train	Valid	Test	All
CMU-MOSI	1284	229	686	2199
CMU-MOEI	16326	1871	4659	22856
CH-SIMS	1368	456	457	2281

CMU-MOSI数据集是一个用于情感分析和情感强度预测的多模态数据集。该数据集主要包含从2199个YouTube电影评论视频中提取的93个简短独白视频片段，涵盖面部表情、语音、文本和上下文信息。每个视频片段都标注有情感标签，范围从-3（高度负面）到3（高度正面），用于表示情感倾向和强度。

CMU-MOSEI：这是CMU-MOSI的增强版本，显著扩展了其规模和范围。该数据集包含22856个带标注的视频片段，涵盖250个不同主题。此数据集与CMU-MOSI数据集具有相同的标签值。

CH-SIMS：CH-SIMS数据集是一个中文多模态情感分析数据集，共收集了来自电影、电视剧和综艺节目的60个原始视频。这些视频被分割成2281个视频片段，为每个模态提供了全面的多模态标注及独立标注。数据集集中的每个样本均被赋予一个从-1（表示强烈负面）到1（表示强烈正面）的情感得分。

4.2 基线方法

To comprehensively validate the performance and effectiveness of the proposed model, we compared our model with several baselines, including some classic models and the most recent state-of-the-art models. The models we compared are as follows:

MuT [19] (2019): Multimodal Transformer (MuT) utilizes directed pairwise cross modal attention to capture correlations between different modalities, enhance interactions between different modalities, and obtain richer multimodal feature representations.

ICCN [43] (2019): The Interaction Canonical Correlation Network (ICCN) learns correlations between all three modes via deep canonical correlation analysis (DCCA) to achieve deep correlations between text and audio, as well as text and video.

MAG-BERT [44] (2020): This model proposes to combine the core architecture of Bert and XLNet [45] with multimodal adaptation gates, and use non-linguistic modal information to adjust language modalities.

MISA [9] (2020): The Modal Invariant and Modal Specific Representation (MISA) divides each modality into modal invariant and modal specific subspaces and learns the modal representations of each subspace, providing a comprehensive perspective for multimodal data.

Self-MM [25] (2021): The model Utilizes self-supervised multi task learning to obtain single modal labels, achieving joint training of multimodal and single modal tasks, helps to learn consistency and differences between modalities.

MMIM [46] (2021): Multimodal Information Modulation (MMIM) is the first to introduce information maximization into multimodal sentiment analysis tasks, enhancing mutual information between input single modal pairs and between multimodal fusion results and single modal inputs through a hierarchical mutual information maximization framework.

HYCON [47] (2023): This method proposes a novel multimodal representation learning framework based on contrastive learning, which designs three types of losses to comprehensively learn inter-modal and intra-modal dynamics in both supervised and unsupervised ways.

SIMR [48] (2023): The speaker independent multimodal representation (SIMR) framework decomposes nonverbal representations into personal style encoding and emotional representation to mitigate the influence of personal style, and uses enhanced cross modal converters to explore contextual interactions between modalities.

MUTA-Net 错误!未找到引用源。 (2023): This modal discourse temporal attention network applies discourse level representation to interactions between different modalities, minimizing intra class distance and maximizing inter class distance, enhancing the discriminative power of representation.

为了全面验证所提出模型的性能和有效性，我们将我们的模型与多个基线进行了比较，包括一些经典模型以及最新的最先进模型。我们比较的模型如下：

MuT [19] (2019): 多模态 Transformer (MuT) 利用定向成对跨模态注意力来捕捉不同模态之间的相关性，增强不同模态间的交互，从而获得更丰富的多模态特征表示。

ICCN [43] (2019)：交互典型相关网络 (ICCN) 通过深度典型相关分析 (DCCA) 学习所有三种模态之间的相关性，以实现文本与音频、文本与视频之间的深度关联。

MAG-BERT [44] (2020)：该模型提出将Bert与XLNet [45]的核心架构与多模态适配门相结合，并利用非语言模态信息来调整语言模态。

MISA [9] (2020): 模态不变与模态特定表示 (MISA) 将每种模态划分为模态不变子空间和模态特定子空间，并学习各子空间的模态表示，为多模态数据提供了全面的视角。

Self-MM [25] (2021)：该模型利用自监督多任务学习来获取单模态标签，实现了多模态与单模态任务的联合训练，有助于学习模态间的一致性与差异。

MMIM [46] (2021年)：多模态信息调制 (MMIM) 首次将信息最大化引入多模态情感分析任务，通过分层互信息最大化框架，增强输入单模态对之间以及多模态融合结果与单模态输入之间的互信息。

HYCON [47] (2023)：该方法提出了一种基于对比学习的多模态表示学习框架，设计了三种损失函数，以有监督和无监督的方式全面学习模态间与模态内的动态关系。

SIMR [48] (2023年)：说话人无关的多模态表示 (SIMR) 框架将非语言表示分解为个人风格编码和情感表示，以减轻个人风格的影响，并利用增强的跨模态转换器来探索模态间的上下文交互。

MUTA-Net (2023): 该模态话语时序注意力网络将话语级表征应用于不同模态间的交互，通过最小化类内距离与最大化类间距离，增强了表征的判别力。

TETFN [22] (2023): The Text Enhanced Transformer Fusion Network (TETFN) obtains effective unified multimodal representation by learning paired cross modal mapping based on text, and pays attention to modal differences through single modal labels.

AOBERT (Kim& Park, 2023) All modalities in One BERT network integrates three types of modal representations using a single stream Transformer and trains MLM and AP tasks using multimodal representations.

ALMT [43] (2023): Adaptive Language Guided Multimodal Converter (ALMT) introduces an Adaptive Hyper modal Learning (AHL) module to learn irrelevant and conflicting information contained in visual and auditory features under the guidance of language features, in order to obtain complementary and joint multimodal representations.

DMD (2023): Decoupling Multimodal Distillation (DMD) preprocessing and extracting features of different modal information, utilizing knowledge distillation techniques to transfer and fuse knowledge between different modalities

MCR [47] (2023): The MCR method draws inspiration from human perceptual paradigms and proposes a collaborative reinforcement method for target and source patterns. This method aims to promote effective cross modal interaction and fusion across various levels of detail.

TMBL [41] (2024): A Transformer based multimodal binding and learning model (TMBL) combines bimodal and tri modal binding mechanisms, fine-grained convolution modules, and utilizes similarity and dissimilarity losses to promote model convergence.

4.3 Experimental parameters

We conducted extensive experiments to investigate the effects of hyperparameters and learning strategies to determine the optimal settings. We adopt the Adam optimizer (Kingma & Ba, 2015) with an initial learning rate of 5×10^{-5} to optimize trainable parameters. Considering computing resources and training efficiency, we have set the batch size to 32. Especially, the early stopping strategy is adopted during the training process, which indicates if the validation loss does not decrease for 8 consecutive epochs, the training process will stop. The model proposed in this article is based on Python 3.9 and PyTorch 1.4.0 implementation. All experiments were conducted using a single GeForce RTX 4090 (24G) GPU under the Windows 10 operating system. The pre trained BERT model can be obtained from the Transformer toolkit published by Hugging Face. The hyperparameter settings during the training process are shown in Table 2.

4.4 Evaluation metrics

TETFN [22] (2023)：文本增强型Transformer融合网络（TETFN）通过基于文本学习成对的跨模态映射，获得有效的统一多模态表示，并借助单模态标签关注模态差异。

AOBERT (Kim & Park, 2023) 在单一BERT网络中整合所有模态，通过单流Transformer集成三种类型的模态表示，并利用多模态表示训练MMLM和AP任务。

ALMT [43] (2023)：自适应语言引导多模态转换器（ALMT）引入了自适应超模态学习（AHL）模块，旨在语言特征的引导下学习视觉与听觉特征中包含的无关及冲突信息，从而获得互补且联合的多模态表征。

DMD (2023)：解耦多模态蒸馏（DMD）通过预处理和提取不同模态信息的特征，利用知识蒸馏技术在不同模态间传递和融合知识

MCR [47] (2023): MCR 方法从人类感知范式中汲取灵感，提出了一种针对目标与源模式的协作强化方法。该方法旨在促进跨模态在不同细节层次上进行有效的交互与融合。

TMBL [41] (2024)：一种基于Transformer的多模态绑定与学习模型（TMBL）融合了双模态与三模态绑定机制、细粒度卷积模块，并利用相似性与相异性损失来促进模型收敛。

4.3 实验参数

我们进行了大量实验以研究超参数和学习策略的影响，从而确定最优设置。我们采用Adam优化器（Kingma & Ba, 2015），初始学习率为 $5^e - 5$ 来优化可训练参数。考虑到计算资源和训练效率，我们将批量大小设置为32。特别地，训练过程中采用了早停策略，即若验证损失连续8个轮次未下降，则终止训练过程。本文提出的模型基于Python 3.9和Python 1.4.0实现。所有实验均在Windows 10操作系统下使用单张GeForce RTX 4090（24G）GPU完成。预训练的BERT模型可从Hugging Face发布的Transformer工具包中获取。训练过程中的超参数设置如表2所示。

4.4 评估指标

This study uses five different metrics to measure the experimental performance of the proposed model: MAE (mean absolute error) directly calculates the error between the true label and the model's predicted label; Corr (Pearson correlation coefficient) measures the degree of prediction bias. Acc7 (seven classification accuracies: sentiment labels between [-3,3]) is used for sentiment rating classification tasks; Acc2 (binary classification accuracy: positive/negative sentiment) and F1 score are used for binary sentiment classification tasks. Except for MAE, higher metric values indicate better model performance. The calculation formulas for these metrics are as follows:

5. Results and analysis

5.1 Quantitative analysis

Table 2 Performance comparison with previous models on CMU-MOSEI. “-”: not presented in the original paper. All results are reported in the original papers. And the results with * are reproduced under the same conditions. The top three results are highlighted in boldface, and the best results are additionally underlined. In Acc-2 and F1-Score, the left of the “/” is calculated as “negative/non-negative” and the right is calculated as “negative/positive”.

Models	MAE (↓)	Acc-2 (↑)	F1-score(↑)	Acc-7 (↑)	Corr (↑)
MuT o	0.871	- / 83.0	- / 82.8	40.0	0.698
ICCN o	0.862	- / 83.07	- / 83.02	39.01	0.714
MAG-BERT*	0.727	82.37 / 84.43	82.50 / 84.61	43.62	0.781
MISA o	0.783	81.8 / 83.4	81.7 / 83.6	42.3	0.761
Self-MM o	0.713	84.0 / 85.98	84.42 / 85.95	-	0.798
MMIM*	0.755	83.67 / 85.37	83.62 / 85.37	45.34	0.773
HYCON o	0.713	- / 85.2	- / 85.1	46.6	0.790
SIMR o	0.706	84.2 / 86.1	84.0 / 86.1	48.2	0.798
MUTA-Net o	0.730	83.1 / 85.0	83.0 / 85.0	-	0.793
TETFN o (2023)	0.717	84.05/86.10	83.83/86.07	-	0.800
AOBERT o	0.856	85.2 / 85.6	85.4 / 86.4	40.2	0.700
ALMT*	0.720	82.94 / 84.91	83.07 / 84.97	47.08	0.783
DMD*	0.736	- / 85.4	- / 85.4	46.34	0.784
MCR o	0.824	- / 84.8	- / 84.6	42.6	0.735
TMBL o	0.867	81.78/83.84	82.41/84.29	36.3	0.762
Ours	0.695	84.84/86.69	84.83/86.72	48.52	0.810

Table 3 Performance comparison with previous models on CMU-MOSEI. “-”: not presented in the

本研究采用五种不同的指标来衡量所提出模型的实验性能：MAE（平均绝对误差）直接计算真实标签与模型预测标签之间的误差；Corr（皮尔逊相关系数）衡量预测偏差的程度。Acc7（七分类准确率：情感标签介于[-3,3]之间）用于情感评分分类任务；Acc2（二分类准确率：正面/负面情感）和F1分数用于二分类情感分类任务。除MAE外，其他指标值越高表示模型性能越好。这些指标的计算公式如下：

5. 结果与分析

5.1 定量分析

表 2 在 CMU-MOSEI 上与先前模型的性能比较。“-”：原始论文中未提供。所有结果均摘自原始论文。带 * 的结果为在相同条件下复现所得。前三名结果以粗体标出，最佳结果额外添加下划线。在 Acc-2 与 F1-Score 中，“/”左侧按“负面/非负面”计算，右侧按“负面/正面”计算。

Models	MAE (↓)	Acc-2 (↑)	F1-score(↑)	Acc-7 (↑)	Corr (↑)
MuT o	0.871	- / 83.0	- / 82.8	40.0	0.698
ICCN o	0.862	- / 83.07	- / 83.02	39.01	0.714
MAG-BERT*	0.727	82.37 / 84.43	82.50 / 84.61	43.62	0.781
MISA o	0.783	81.8 / 83.4	81.7 / 83.6	42.3	0.761
Self-MM o	0.713	84.0 / 85.98	84.42 / 85.95	-	0.798
MMIM*	0.755	83.67 / 85.37	83.62 / 85.37	45.34	0.773
HYCON o	0.713	- / 85.2	- / 85.1	46.6	0.790
SIMR o	0.706	84.2 / 86.1	84.0 / 86.1	48.2	0.798
MUTA-Net o	0.730	83.1 / 85.0	83.0 / 85.0	-	0.793
TETFN o (2023)	0.717	84.05/86.10	83.83/86.07	-	0.800
AOBERT o	0.856	85.2 / 85.6	85.4 / 86.4	40.2	0.700
ALMT*	0.720	82.94 / 84.91	83.07 / 84.97	47.08	0.783
DMD*	0.736	- / 85.4	- / 85.4	46.34	0.784
MCR o	0.824	- / 84.8	- / 84.6	42.6	0.735
TMBL o	0.867	81.78/83.84	82.41/84.29	36.3	0.762
Ours	0.695	84.84/86.69	84.83/86.72	48.52	0.810

表3 在CMU-MOSEI上与先前模型的性能比较。“-”：未在

original paper. All results are reported in the original papers. And the results with * are reproduced under the same conditions. The top three results are highlighted in boldface, and the best results are additionally underlined. In Acc-2 and F1-Score, the left of the “/” is calculated as “negative/non-negative” and the right is calculated as “negative/positive”.

Models	MAE (↓)	Acc-2 (↑)	F1-score(↑)	Acc-7 (↑)	Corr (↑)
MuT o	0.580	- / 82.5	- / 82.3	51.8	0.703
ICCN o	0.565	- / 84.18	- / 84.15	51.58	0.713
MAG-BERT*	0.543	82.5 / 84.82	82.77 / 84.71	52.67	0.755
MISA o	0.555	83.6 / 85.5	83.8 / 85.3	52.2	0.756
Self-MM o	0.530	82.81 / 85.17	82.53 / 85.30	-	0.765
MMIM*	0.543	83.08 / 85.04	83.48 / 84.88	45.34	0.758
HYCON o	0.601	- / 85.4	- / 85.6	52.8	0.776
SIMR o	0.580	82.5 / 82.9	81.9 / 82.9	52.6	0.696
MUTA-Net o	0.544	82.4 / 85.0	82.7 / 84.9	-	0.760
TETFN o (2023)	0.551	84.25 / 85.18	84.18 / 85.27	-	0.748
AOBERT o	0.515	84.9 / 86.2	85.0 / 85.9	54.5	0.763
ALMT*	0.533	82.23 / 85.66	81.89 / 85.79	52.82	0.777
DMD*	0.541	- / 85.5	- / 85.4	52.5	0.762
MCR o	0.554	- / 84.7	- / 84.3	54.5	0.736
TMBL o	0.545	84.23/85.84	84.87/85.92	52.4	0.766
Ours	0.526	83.69/85.97	83.86/85.98	54.8	0.768

5.2 Ablation study

Conclusion

This paper proposes a multimodal sentiment analysis model with a hybrid of three fusion methods for multimodal sentiment analysis tasks. This model combines the advantages of early fusion and model level fusion, and obtaining strong inter-modal connection features between modalities through multi type and multi-level fusion. Extensive experiments on CMU-MOSI and CMU-MOSEI datasets have shown that it has advantages over other methods in multimodal sentiment analysis tasks. Empirically, textual features are usually superior to non-textual features. In

原始论文。所有结果均报告于原始论文中。带 * 的结果是在相同条件下复现的。前三名结果以**粗体**突出显示，最佳结果额外添加下划线。在 Acc-2 和 F1-Score 中，“/” 左侧按 “负面/非负面” 计算，右侧按 “负面/正面” 计算。

Models	MAE (↓)	Acc-2 (↑)	F1-score(↑)	Acc-7 (↑)	Corr (↑)
MuT o	0.580	- / 82.5	- / 82.3	51.8	0.703
ICCN o	0.565	- / 84.18	- / 84.15	51.58	0.713
MAG-BERT*	0.543	82.5 / 84.82	82.77 / 84.71	52.67	0.755
MISA o	0.555	83.6 / 85.5	83.8 / 85.3	52.2	0.756
Self-MM o	0.530	82.81 / 85.17	82.53 / 85.30	-	0.765
MMIM*	0.543	83.08 / 85.04	83.48 / 84.88	45.34	0.758
HYCON o	0.601	- / 85.4	- / 85.6	52.8	0.776
SIMR o	0.580	82.5 / 82.9	81.9 / 82.9	52.6	0.696
MUTA-Net o	0.544	82.4 / 85.0	82.7 / 84.9	-	0.760
TETFN o (2023)	0.551	84.25 / 85.18	84.18 / 85.27	-	0.748
AOBERT o	0.515	84.9 / 86.2	85.0 / 85.9	54.5	0.763
ALMT*	0.533	82.23 / 85.66	81.89 / 85.79	52.82	0.777
DMD*	0.541	- / 85.5	- / 85.4	52.5	0.762
MCR o	0.554	- / 84.7	- / 84.3	54.5	0.736
TMBL o	0.545	84.23/85.84	84.87/85.92	52.4	0.766
Ours	0.526	83.69/85.97	83.86/85.98	54.8	0.768

5.2 消融研究

结论

本文提出了一种多模态情感分析模型，该模型融合了三种融合方法，适用于多模态情感分析任务。该模型结合了早期融合和模型级融合的优势，通过多类型、多层次的融合方式，获取模态间强大的跨模态关联特征。在CMU-MOSI和CMU-MOSEI数据集上的大量实验表明，该模型在多模态情感分析任务中优于其他方法。经验表明，文本特征通常优于非文本特征。

future work, we will further explore the contribution of strong and weak features in multimodal sentiment analysis tasks.

Competing Interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Authors contribution statement

Liang Yang: Conceptualization, Methodology, Software, Formal Analysis, Writing - Original Draft. **Xinhua Zhu:** Data Curation, Supervision, Funding Acquisition, Writing - Review & Editing. **Siyang Li:** Investigation, Resources, Project Administration.

Ethical and informed consent for data used

The datasets used in this study were CMU-MOSI and CMU-MOSEI created by researchers from Carnegie Mellon University. These datasets are publicly accessible and designed for academic use in the field of multimodal sentiment analysis. They do not contain any personally identifiable information or sensitive data. The use of these datasets complies with the terms and conditions set forth by their respective publishers. No specific informed consent was required for this study as the data used were obtained from publicly accessible sources which do not contain information related to individual human subjects. These data sources are open for research purposes and are anonymized in such a manner that individual privacy is maintained.

Data availability and access

The datasets used and analysed during the current study are available from the corresponding author.

Acknowledgements

This work was supported in part by the National Natural Science Foundation of China under Contract 62062012 and Guangxi Collaborative Innovation Center of Multi-source Information Integration and Intelligent Processing.

Reference

- [1] Kaur, R. & Kautish, S. (2019). Multimodal Sentiment Analysis: A Survey and Comparison. *International Journal of Service Science, Management, Engineering, and Technology (IJSSMET)*, 10(2), 38-58.
<http://doi.org/10.4018/IJSSMET.2019040103>.
- [2] Zhu, L., Zhu, Z., Zhang, C., Xu, Y., & Kong, X. (2023). Multimodal sentiment analysis based on fusion methods: a survey. *Information Fusion*.
- [3] Soleymani, Mohammad, Garcia, David, Jou, & Brendan, et al. (2017). A survey of multimodal sentiment analysis. *Image & Vision Computing*.
<http://doi.org/10.1016/j.imavis.2017.08.003>.

未来的工作中，我们将进一步探索强特征和弱特征在多模态情感分析任务中的贡献。

利益冲突

作者声明，对于本论文所报道的工作，不存在任何已知的可能影响研究结果的竞争性经济利益或个人关系。

作者贡献声明

梁阳：概念化、方法论、软件、形式分析、初稿撰写。朱新华：数据管理、监督、资金获取、文稿审阅与编辑。李思洋：调查、资源、项目管理。

数据使用的伦理和知情同意

本研究使用的数据集为卡内基梅隆大学研究人员创建的CMU-MOSI与CMU-MOSEI。这些数据集公开可用，专为多模态情感分析领域的学术研究设计，不包含任何个人身份信息或敏感数据。其使用符合相应发布方规定的条款与条件。由于数据来源于不涉及人类受试者信息的公开渠道，本研究无需获取特定知情同意。这些数据源面向研究目的开放，且经过匿名化处理以充分保障个人隐私。

数据可用性与访问

当前研究中使用和分析的数据集可从通讯作者处获取。

致谢

本工作部分得到了国家自然科学基金（项目编号：62062012）和广西多源信息集成与智能处理协同创新中心的支持。

参考

[1] Kaur, R. & Kautish, S. (2019). 多模态情感分析：综述与比较. *International Journal of Service Science, Management, Engineering, and Technology (IJSSMET)*, 10(2), 38-58. <http://doi.org/10.4018/IJSSMET.2019040103>.

[2] Zhu, L., Zhu, Z., Zhang, C., Xu, Y., & Kong, X. (2023). 基于融合方法的多模态情感分析综述. *Information Fusion*. [3] Soleymani, Mohammad, Garcia, David, Jou, & Brendan, et al. (2017). 多模态情感分析综述. *Image & Vision Computing*. <http://doi.org/10.1016/j.imavis.2017.08.003>.

- [4] Bakshi, R. K., Kaur, N., Kaur, R., & Kaur, G. (2016). Opinion mining and sentiment analysis. Department of Computer Science and Engineering, Guru Tegh Bahadur Institute of Technology, G-8 Area, Rajouri Garden, New Delhi-110064, India.
- [5] Guo, X., Yu, W., & Wang, X. (2021). An overview on fine-grained text sentiment analysis: survey and challenges. *Journal of Physics Conference Series*, 1757(1), 012038.
<http://doi.org/10.1088/1742-6596/1757/1/012038>.
- [6] Zhang, L., Wang, S., & Liu, B. (2018). Deep learning for sentiment analysis: a survey. *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery*, e1253.
<http://doi.org/10.1002/widm.1253>
- [7] Zadeh, A., Chen, M., Poria, S., Cambria, E., & Morency, L. P. (2017). Tensor Fusion Network for Multimodal Sentiment Analysis, in: *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing*.
<http://doi.org/10.18653/v1/D17-1115>.
- [8] Liu, Z., Shen, Y., Lakshminarasimhan, V. B., Liang, P. P., Zadeh, A., & Morency, L. P. (2018). Efficient low-rank multimodal fusion with modality-specific factors.
<http://doi.org/10.18653/v1/P18-1209>
- [9] Hazarika, D., Zimmermann, R., & Poria, S. (2020). Misa: modality-invariant and -specific representations for multimodal sentiment analysis, in: *Proceedings of the 28th ACM International Conference on Multimedia*, 2020, pp. 1122-1131.
- [10] Zhang, L., Jin, L., Xu, G., Li, X., Xu, C., Wei, K., Liu, N., & Liu, H. (2024). CAMEL: Capturing Metaphorical Alignment with Context Disentangling for Multimodal Emotion Recognition. *Proceedings of the AAAI Conference on Artificial Intelligence*, 38(8), 9341-9349.
<https://doi.org/10.1609/aaai.v38i8.28787>
- [11] S. Bai, J. Z. Kolter, V. Koltun. (2018). An Empirical Evaluation of Generic Convolutional and Recurrent Networks for Sequence Modeling.
<https://doi.org/10.48550/arXiv.1803.01271>
- [12] Vaswani, A., Shazeer, N. M., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, L., & Polosukhin, I. (2017). Attention is All you Need. *Neural Information Processing Systems*.
- [13] Chandrasekaran G, Nguyen T N, Jude H D. Multimodal sentimental analysis for social media applications: A comprehensive review. *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery* [2024-05-11].
<https://10.1002/widm.1415>.

[4] Bakshi, R. K., Kaur, N., Kaur, R., & Kaur, G. (2016). 观点挖掘与情感分析。计算机科学与工程系, Guru Tegh Bahadur 技术学院, G-8 区, Rajouri Garden, 新德里-110064, 印度。 [5] Guo, X., Yu, W., & Wang, X. (2021). 细粒度文本情感分析综述: 调查与挑战。物理学报会议系列, 1757(1), 012038。 <http://doi.org/10.1088/1742-6596/1757/1/012038>. [6] Zhang, L., Wang, S., & Liu, B. (2018). 深度学习用于情感分析: 综述。Wiley 跨学科评论: 数据挖掘与知识发现, e1253。 <http://doi.org/10.1002/widm.1253> [7] Zadeh, A., Chen, M., Poria, S., Cambria, E., & Morency, L. P. (2017). 用于多模态情感分析的张量融合网络, 收录于: 2017年自然语言处理经验方法会议论文集。 <http://doi.org/10.18653/v1/D17-1115>.

[8] 刘, Z., 沈, Y., Lakshminarasimhan, V. B., 梁, P. P., Zadeh, A., & Morency, L. P. (2018). 使用模态特定因子的高效低秩多模态融合。 <http://doi.org/10.18653/v1/P18-1209>

[9] Hazarika, D., Zimmermann, R., & Poria, S. (2020). MISA: 面向多模态情感分析的模态不变与模态特定表示。载于: 第28届ACM国际多媒体会议论文集, 2020年, 第1122-1131页。 [10] Zhang, L., Jin, L., Xu, G., Li, X., Xu, C., Wei, K., Liu, N., & Liu, H. (2024). CAMEL: 通过上下文解耦捕获隐喻对齐以实现多模态情绪识别。AAAI人工智能会议论文集, 38(8), 9341-9349. <https://doi.org/10.1609/aaai.v38i8.28787> [11] S. Bai, J. Z. Kolter, V. Koltun. (2018). 通用卷积与循环网络用于序列建模的实证评估。 <https://doi.org/10.48550/arXiv.1803.01271> [12] Vaswani, A., Shazeer, N. M., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, L., & Polosukhin, I. (2017). Attention Is All You Need. 神经信息处理系统。 [13] Chandrasekaran G, Nguyen T N, Jude H D. 面向社交媒体应用的多模态情感分析: 一项全面综述。Wiley 跨学科评论: 数据挖掘与知识发现 [2024-05-11]. <https://10.1002/widm.1415>.

- [14] Ananya Pandey, Dinesh Kumar Vishwakarma. Progress, achievements, and challenges in multimodal sentiment analysis using deep learning: A survey. *Applied Soft Computing*, Volume 152, 2024, 111206.
<https://doi.org/10.1016/j.asoc.2023.111206>
- [15] Zhu, T., Li, L., Yang, J., Zhao, S., Liu, H. et al. (2022). Multimodal sentiment analysis with image-text interaction network. *IEEE Transactions on Multimedia*, 25, 3375–3385.
- [16] Lai S., Hu X., Xu H., Ren Z., & Liu Z. (2023). Multimodal sentiment analysis: A survey, *Displays*, Volume 80, 2023, 102563, ISSN 0141-9382,
<https://doi.org/10.1016/j.displa.2023.102563>.
- [17] Malhotra, P., Vig, L., Shroff, G., & Agarwal, P. (2015). Long Short-Term Memory Networks for Anomaly Detection in Time Series. 23rd European Symposium on Artificial Neural Networks, Computational Intelligence and Machine Learning, ESANN 2015.
- [18] Wang, Y., Shen, Y., Liu, Z., Liang, P. P., Zadeh, A., & Morency, L.-P. (2019). Words Can Shift: Dynamically Adjusting Word Representations Using Nonverbal Behaviors. *Proceedings of the AAAI Conference on Artificial Intelligence*, 33(01), 7216-7223.
<https://doi.org/10.1609/aaai.v33i01.33017216>
- [19] Tsai, Y. H. H., Bai, S., Liang, P. P., Kolter, J. Z., & Salakhutdinov, R. (2019). Multimodal transformer for unaligned multimodal language sequences, in: *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 6558–6569.
<https://doi.org/10.18653/v1/P19-1656>
- [20] Wang, F., Tian, S., Yu, L. et al. TEDT: Transformer-Based Encoding–Decoding Translation Network for Multimodal Sentiment Analysis. *Cogn Comput* 15, 289–303 (2023).
<https://doi.org/10.1007/s12559-022-10073-9>
- [21] Zhang, Q., Shi, L., Liu, P., Zhu, Z., & Xu, L. (2023). ICDN: integrating consistency and difference networks by transformer for multimodal sentiment analysis. *Applied Intelligence*.
<https://doi.org/10.1007/s10489-023-04869-x>
- [22] Wang D., Guo X., Tian Y., Liu J., He L., & Luo X. (2023). TETFN: A text enhanced transformer fusion network for multimodal sentiment analysis. *Pattern Recognition*, Volume 136, 2023, 109259, ISSN 0031-3203.
<https://doi.org/10.1016/j.patcog.2022.109259>.
- [23] Fu Y., Huang B., Wen Y., & Zhang P. (2024). FDR-MSA: Enhancing multimodal sentiment analysis through feature disentanglement and

[14] Ananya Pandey, Dinesh Kumar Vishwakarma. 基于深度学习的多模态情感分析进展、成就与挑战：综述. *Applied Soft Computing*, Volume 152, 2024, 111206. <http://doi.org/10.1016/j.asoc.2023.111206>

[15] Zhu, T., Li, L., Yang, J., Zhao, S., Liu, H. 等人. (2022). 基于图文交互网络的多模态情感分析. *IEEE Transactions on Multimedia*, 25, 3375–3385.

[16] Lai S., Hu X., Xu H., Ren Z., & Liu Z. (2023). 多模态情感分析：综述, 《Displays》, 第80卷, 2023年, 102563页, ISSN 0141-9382, <https://doi.org/10.1016/j.displa.2023.102563>.

[17] Malhotra, P., Vig, L., Shroff, G., & Agarwal, P. (2015). 用于时间序列异常检测的长短期记忆网络. 第23届欧洲人工神经网络、计算智能与机器学习研讨会, E SANN 2015. [18] Wang, Y., Shen, Y., Liu, Z., Liang, P. P., Zadeh, A., & Morency, L.-P. (2019). 词语可迁移：利用非言语行为动态调整词表示. *AAAI人工智能会议论文集*, 33(01), 7216-7223. <https://doi.org/10.1609/aaai.v33i01.33017216>

[19] Tsai, Y. H. H., Bai, S., Liang, P. P., Kolter, J. Z., & Salakhutdinov, R. (2019). 用于非对齐多模态语言序列的多模态Transformer, 收录于：第57届计算语言学协会年会论文集, 第6558–6569页. <https://doi.org/10.18653/v1/P19-1656>

[20] Wang, F., Tian, S., Yu, L. 等. TEDT：基于Transformer编码-解码翻译网络的多模态情感分析. *Cogn Comput* 15, 289–303 (2023). <https://doi.org/10.1007/s12559-022-10073-9>

[21] 张, Q., 石, L., 刘, P., 朱, Z., & 徐, L. (2023). ICDN：基于Transformer融合一致性与差异性网络的多模态情感分析. *Applied Intelligence*. <https://doi.org/10.1007/s10489-023-04869-x>

[22] 王 D., 郭 X., 田 Y., 刘 J., 何 L., & 罗 X. (2023). TETFN：一种用于多模态情感分析的文本增强型Transformer融合网络. *模式识别*, 第136卷, 2023年, 109259, ISSN 0031-3203. <https://doi.org/10.1016/j.patcog.2022.109259>.

[23] 傅 Y., 黄 B., 温 Y., & 张 P. (2024). FDR-MSA：通过特征解耦与

reconstruction. Knowledge-Based Systems, Volume 297, 2024, 111965, ISSN 0950-7051.

<https://doi.org/10.1016/j.knosys.2024.111965>.

- [24] Li, Y., Zhang, K., Wang, J., & Gao, X. (2020). A cognitive brain model for multimodal sentiment analysis based on attention neural networks. *Neurocomputing*, 430(2).

<https://doi.org/10.1016/j.neucom.2020.10.021>

- [25] Yu, W., Xu, H., Yuan, Z., & Wu, J. (2021). Learning Modality-Specific Representations with Self-Supervised Multi-Task Learning for Multimodal Sentiment Analysis. *Proceedings of the AAAI Conference on Artificial Intelligence*, 35(12), 10790-10797.

<https://doi.org/10.1609/aaai.v35i12.17289>

- [26] Shutova, E., Kiela, D., & Maillard, J. (2016). Black Holes and White Rabbits: Metaphor Identification with Visual Features. *Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*.

<https://doi.org/10.18653/v1/N16-1020>

- [27] Morvant, E., Habrard, A., Ayache, S. (2014). Majority Vote of Diverse Classifiers for Late Fusion. In: Fränti, P., Brown, G., Loog, M., Escolano, F., Pelillo, M. (eds) *Structural, Syntactic, and Statistical Pattern Recognition. S+SSPR 2014. Lecture Notes in Computer Science*, vol 8621. Springer, Berlin, Heidelberg.

https://doi.org/10.1007/978-3-662-44415-3_16

- [28] Evangelopoulos, G., Zlatintsi, A., Potamianos, A., Maragos, P. A., Rapantzikos, K. E., & Skoumas, G., et al. (2013). Multimodal saliency and fusion for movie summarization based on aural, visual, and textual attention. *IEEE Transactions on Multimedia*, pp.1553-1568

- [29] Zadeh, A., Liang, P. P., Poria, S., Vij, P., & Morency, L. P. (2018). Multi-attention recurrent network for human communication comprehension. *Proceedings of the AAAI Conference on Artificial Intelligence*, 2018.

<https://doi.org/10.48550/arXiv.1802.00923>

- [30] Ou Y., Chen Z., & Wu F. (2021) Multimodal Local-Global Attention Network for Affective Video Content Analysis, in *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 31, no. 5, pp. 1901-1914, May 2021.

<https://doi.org/10.1109/TCSVT.2020.3014889>

- [31] Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2018). Bert: pre-training of deep bidirectional transformers for language understanding, in: *Proceedings of NaacL-HLT*, Vol. 1, 2019, p. 2.

重建。基于知识的系统, 第297卷, 2024年, 111965, ISSN 0950-7051。 <https://doi.org/10.1016/j.knosys.2024.111965>。

[24] 李, Y., 张, K., 王, J., & 高, X. (2020). 一种基于注意力神经网络的认知脑模型用于多模态情感分析. 神经计算, 430(2). <https://doi.org/10.1016/j.neucom.2020.10.021>[25] 余, W., 徐, H., 袁, Z., & 吴, J. (2021). 通过自监督多任务学习学习模态特定表示以进行多模态情感分析. AAAI人工智能会议论文集, 35(12), 1079 0-10797. <https://doi.org/10.1609/aaai.v35i12.17289>

[26] Shutova, E., Kiela, D., & Maillard, J. (2016). 黑洞与白兔: 利用视觉特征进行隐喻识别. 北美计算语言学协会人类语言技术分会会议. <https://doi.org/10.18653/v1/N16-1020>

[27] Morvant, E., Habrard, A., Ayache, S. (2014). 用于晚期融合的多样化分类器多数投票. 载于: Fränti, P., Brown, G., Loog, M., Escolano, F., Pelillo, M. (编) 结构、句法与统计模式识别. S+SSPR 2014. 计算机科学讲义, 卷 8621. Springer, Berlin, Heidelberg. https://doi.org/10.1007/978-3-662-44415-3_16[28] Evangelopoulos, G., Zlatintsi, A., Potamianos, A., Maragos, P. A., Rapantzikos, K. E., & Skoumas, G., 等. (2013). 基于听觉、视觉和文本注意力的多模态显著性与融合用于电影摘要生成. IEEE多媒体汇刊, 第1553-1568页。[29] Zadeh, A., Liang, P. P., Poria, S., Vij, P., & Morency, L. P. (2018). 用于人类交流理解的多注意力循环网络. 人工智能AAAI会议论文集, 2018. <https://doi.org/10.48550/arXiv.1802.00923>[30] Ou Y., Chen Z., & Wu F. (2021). 用于情感视频内容分析的多模态局部-全局注意力网络, 载于IEEE电路与系统视频技术汇刊, 卷31, 第5期, 第1901-1914页, 2021年5月。 <https://doi.org/10.1109/TCSVT.2020.3014889>[31] Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2018). Bert: 用于语言理解的深度双向Transformer预训练, 载于: NaacL-HLT会议论文集, 卷1, 2019年, 第2页。

- [32] Baltrusaitis, T., Zadeh, A., Lim, Y. C., & Morency, L. P. (2018). OpenFace 2.0: Facial Behavior Analysis Toolkit. IEEE International Conference on Automatic Face & Gesture Recognition (pp.59-66). IEEE Computer Society.
- [33] Yu, W., Xu, H., Meng, F., Zhu, Y., & Yang, K. (2020). CH-SIMS: A Chinese Multimodal Sentiment Analysis Dataset with Fine-grained Annotation of Modality. Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics.
<https://doi.org/10.18653/v1/2020.acl-main.343>
- [34] Zadeh, A., Zellers, R., Pincus, E., & Morency, L. P. (2016). Multimodal sentiment intensity analysis in videos: facial gestures and verbal messages. IEEE Intelligent Systems, 31(6), 82-88.
<https://doi.org/10.1109/MIS.2016.94>
- [35] Zadeh A., Liang P.P., Poria S., Cambria E., & Morency L.-P. (2018) Multimodal Language Analysis in the Wild: CMU-MOSEI Dataset and Interpretable Dynamic Fusion Graph, in: Proceedings of the 56th annual meeting of the association for computational linguistics.
<https://doi.org/10.18653/v1/P18-1208>
- [36] Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. Neural Computation, 9(8), 1735-1780.
<https://doi.org/10.1162/neco.1997.9.8.1735>
- [37] Jürgen Schmidhuber. (2015). Deep learning in neural networks: an overview. Neural Networks, 61, 85-117.
- [38] Dalal, N., & Triggs, B. (2005). Histograms of Oriented Gradients for Human Detection. IEEE Computer Society Conference on Computer Vision & Pattern Recognition. IEEE.
- [39] Degottex, G., Kane, J., Drugman, T., Raitio, T., & Scherer, S. (2014). COVAREP: A Collaborative Voice Analysis Repository for Speech Technologies. Acoustics Speech and Signal Processing (ICASSP), 2014 IEEE International Conference on. IEEE.
- [40] Chen L., Zhao Y., Chan J.C.-W., & Kong S.G. (2022) Histograms of oriented mosaic gradients for snapshot spectral image description. ISPRS Journal of Photogrammetry and Remote Sensing, 183 (2022), pp. 79-93.
<https://doi.org/10.1016/j.isprsjprs.2021.10.018>
- [41] Mcfee, B., Raffel, C., Liang, D., Ellis, D., & Nieto, O. (2015). Librosa: Audio and Music Signal Analysis in Python, in: Proceedings of the 14th python in science conference, volume 8.
- [42] Li, R., Wu, Z., Jia, J., Li, J., Chen, W., & Meng, H. (2018). Inferring user emotive state changes in realistic human-computer conversational dialogs. In 2018 ACM Multimedia Conference on Multimedia Conference, pages 136–144. ACM.

[32] Baltrusaitis, T., Zadeh, A., Lim, Y. C., & Morency, L. P. (2018). OpenFace 2.0: 面部行为分析工具包。IEEE 自动人脸与手势识别国际会议 (pp.59-66). IEEE 计算机学会。[33] Yu, W., Xu, H., Meng, F., Zhu, Y., & Yang, K. (2020). CH-SIMS: 一个带有细粒度模态标注的中文多模态情感分析数据集。第58届计算语言学协会年会论文集。 <https://doi.org/10.18653/v1/2020.acl-main.343>[34] Zadeh, A., Zellers, R., Pincus, E., & Morency, L. P. (2016). 视频中的多模态情感强度分析: 面部表情与言语信息。IEEE 智能系统, 31(6), 82-88。 <https://doi.org/10.1109/MIS.2016.94>[35] Zadeh A., Liang P.P., Poria S., Cambria E., & Morency L.-P. (2018) 真实环境下的多模态语言分析: CMU-MOSEI 数据集与可解释动态融合图, 载于: 第56届计算语言学协会年会论文集。 <https://doi.org/10.18653/v1/P18-1208>[36] Hochreiter, S., & Schmidhuber, J. (1997). 长短期记忆。神经计算, 9(8), 1735-1780。 <https://doi.org/10.1162/neco.1997.9.8.1735>[37] Jürgen Schmidhuber. (2015). 神经网络中的深度学习: 综述。神经网络, 61, 85-117。[38] Dalal, N., & Triggs, B. (2005). 用于人体检测的定向梯度直方图。IEEE 计算机学会计算机视觉与模式识别会议。IEEE。[39] Degottex, G., Kane, J., Drugman, T., Raitio, T., & Scherer, S. (2014). COVAR EP: 一个用于语音技术的协作语音分析库。2014年IEEE声学、语音与信号处理国际会议。IEEE。[40] Chen L., Zhao Y., Chan J.C.-W., & Kong S.G. (2022) 用于快照光谱图像描述的定向马赛克梯度直方图。ISPRS 摄影测量与遥感杂志, 183 (2022), pp. 79-93。 <https://doi.org/10.1016/j.isprsjprs.2021.10.018>[41] Mcfee, B., Raffel, C., Liang, D., Ellis, D., & Nieto, O. (2015). Librosa: Python中的音频与音乐信号分析, 载于: 第14届Python科学会议论文集, 第8卷。[42] Li, R., Wu, Z., Jia, J., Li, J., Chen, W., & Meng, H. (2018). 在真实人机会话对话中推断用户情感状态变化。2018年ACM多媒体会议, 第136-144页。ACM。

- [43] Z. Sun, P. Sarma, W. Sethares, & Y. Liang. (2019). Learning relationships between text, audio, and video via deep canonical correlation for multimodal language analysis, in: Proceedings of the AAAI Conference on Artificial Intelligence, Vol. 1, 2019, pp. 2236–2246.
- [44] W. Rahman, M.K. Hasan, S. Lee, A. Zadeh, C. Mao, L.-P. Morency, E. Hoque, Integrating multimodal information in large pre-trained transformers, in: Proceedings of the Conference. Association for Computational Linguistics. Meeting, 2020, p. 2359. <https://doi.org/10.48550/arXiv.1908.05787>
- [45] Yang, Z., Dai, Z., Yang, Y., Carbonell, J., Salakhutdinov, R., & Le, Q. V. (2019). Xlnet: generalized autoregressive pretraining for language understanding. <https://doi.org/10.48550/arXiv.1906.08237>
- [46] Han, W., Chen, H., & Poria, S. (2021). Improving multimodal fusion with hierarchical mutual information maximization for multimodal sentiment analysis, in: Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing, 2021, pp. 9180–9192. <https://doi.org/10.48550/arXiv.2109.00412>
- [47] Mai, S., Zeng, Y., Zheng, S., & Hu, H. (2023). Hybrid contrastive learning of tri-modal representation for multimodal sentiment analysis. IEEE transactions on affective computing, 14(3), 2276-2289.
- [48] Wang, J., Wang, S., Lin, M., Xu, Z., & Guo, W. (2023). Learning speaker-independent multimodal representation for sentiment analysis. Information Sciences: An International Journal. <https://doi.org/10.1016/j.ins.2023.01.116>
- [49] Tang Z., Xiao Q., Zhou X., Li Y., Chen C., & Li K. (2023). Learning discriminative multi-relation representations for multimodal sentiment analysis. Information Sciences, Volume 641, 2023, 119125, ISSN 0020-0255. <https://doi.org/10.1016/j.ins.2023.119125>.
- [50] s

[43] 孙振、P. Sarma、W. Sethares、梁勇. (2019). 通过深度典型相关学习文本、音频和视频之间的关系以进行多模态语言分析, 收录于: 《AAAI人工智能会议论文集》, 第1卷, 2019年, 第2236–2246页。 [44] W. Rahman, M.K. Hasan, S. Lee, A. Zadeh, C. Mao, L.-P. Morency, E. Hoque, 在大型预训练Transformer中集成多模态信息, 收录于: 《计算语言学协会会议论文集》, 2020年, 第2359页。 <https://doi.org/10.48550/arXiv.1908.05787> [45] Yang, Z., Dai, Z., Yang, Y., Carbonell, J., Salakhutdinov, R., & Le, Q. V. (2019). XLNet: 用于语言理解的广义自回归预训练。 <https://doi.org/10.48550/arXiv.1906.08237> [46] 韩伟、陈浩、Poria, S. (2021). 通过层次化互信息最大化改进多模态融合以进行多模态情感分析, 收录于: 《2021年自然语言处理实证方法会议论文集》, 2021年, 第9180–9192页。 <https://doi.org/10.48550/arXiv.2109.00412> [47] 麦松、曾毅、郑帅、胡海. (2023). 用于多模态情感分析的三模态表征混合对比学习。《IEEE情感计算汇刊》, 14(3), 2276–2289. [48] 王健、王帅、林敏、徐震、郭伟. (2023). 学习说话人无关的多模态表征以进行情感分析。《信息科学: 国际期刊》。 <https://doi.org/10.1016/j.ins.2023.01.116> [49] 唐震、肖强、周星、李阳、陈晨、李科. (2023). 学习判别性多关系表征以进行多模态情感分析。《信息科学》, 第641卷, 2023年, 119125页, ISSN 0020-0255。 <https://doi.org/10.1016/j.ins.2023.119125>. [50] s
